HEALTHCONNECT RWANDA

Week 7 — Data Analytics Testing, Validation & Refinement

AnalystLab Africa Experience Lab | Data Analytics Track | 31 October 2026

Executive summary

The 19 uploaded source files were loaded and profiled. The collection contains 258,375 rows across the loaded tables/sheets (including duplicate representations of some master tables in separate CSV/XLSX files). The patient master contains 20,000 records; the clinical workbook contains 5,000 appointment rows dated 2026-01-01 to 2026-12-28; billing and diagnosis CSVs each contain 60,000 rows. The analysis uncovered material referential-integrity and reconciliation problems. Consequently, descriptive results are reported only within their source-table boundaries, and cross-table conclusions are withheld where joins fail.

Week 6 → Week 7 transition

Week 6 produced the analytics framework, KPI dictionary/validation checklist, SQL query templates and a six-page dashboard wireframe. Week 7 was required to test and refine those outputs. With the actual files now supplied, the work moved from blocked templates to source profiling, relationship checks, independent summaries, and an issue register. The central finding is that the uploaded tables are not currently safe to join comprehensively because several appointment-level keys do not reconcile.

Dataset inventory

29 file/sheet-level datasets were loaded. 258,375 rows were profiled. The inventory workbook records row counts, columns, missing cells and exact duplicate rows for every file/sheet. Patient table missingness is 5,071 cells; other tables shown in this inventory have no missing cells at file level.

Key descriptive results

Patients: 20,000 records, 2 gender categories; female 9,910, male 10,090. The chronic-condition field is blank for 5,071 records; blanks are treated as unknown/not recorded. Clinical appointments: 5,000 rows across 20 facility IDs, with 2026-01-01–2026-12-28 coverage. Visit types: Follow-up: 1,694, Consultation: 1,656, Emergency: 1,650.

Feedback: 21,064 records; average numeric rating 2.99/5. Rating distribution: 1: 4,245, 2: 4,273, 3: 4,202, 4: 4,195, 5: 4,149. Comment labels conflict with ratings and should not be interpreted as sentiment until clarified.

Financial: billed amount total 9,309,401,961; payment amount total 8,389,454,556; separate monthly revenue summary total 1,132,677,500 and appointment count 60,809. These are separate reported aggregates and are not reconciled. Do not label billing totals as recognized revenue or payment totals as cash collected.

Critical data-quality findings

Issue	Severity	Evidence
DQ-01 — Appointment master coverage gap	Critical	5,000 appointment rows in workbook, but billing and diagnosis CSVs contain 60,000 appointment references
DQ-02 — Feedback linkage gap	Critical	19,308 of 21,064 feedback rows (91.7%) reference appointment IDs absent from the 5,000-row appointment
DQ-03 — Facility/department inconsistency	High	4,744 of 5,000 appointment rows have a FacilityID that differs from the facility assigned to DepartmentID.
DQ-04 — Patient table missingness	Medium	5,071 missing cells in patient table; ChronicCondition is blank for 5,071 patients.
DQ-05 — Billing status does not reconcile to payment amount	High	11,887 of 60,000 billing rows have AmountPaid different from Amount, while all three billing statuses occur a
DQ-06 — Revenue summary and transaction totals disagree	Critical	Monthly summary reports 60,809 appointments and revenue 1,132,677,500; the clinical workbook contains c
DQ-07 — All beds marked Available	High	2,000 of 2,000 beds are marked Available; no occupied/maintenance states are present.
DQ-08 — Feedback comment/rating semantic conflict	Medium	Comment labels (Poor/Average/Good/Excellent) do not show a consistent relationship to numeric ratings 1–5

Foreign-key validation

Relationship	Rows	Unmatched	Rate
Appointments.PatientID → Patients.PatientID	5,000	0	0.0%
Appointments.DoctorID → Doctor.StaffID	5,000	0	0.0%
Appointments.DepartmentID → Departments.DepartmentID	5,000	0	0.0%

Relationship	Rows	Unmatched	Rate
Appointments FacilityID agrees with Department FacilityID	5,000	4,744	94.9%
Billing.AppointmentID → Appointments.AppointmentID	60,000	55,000	91.7%
Payments.BillingID → Billing.BillingID	60,000	0	0.0%
Feedback.AppointmentID → Appointments.AppointmentID	21,064	19,308	91.7%
Diagnosis CSV AppointmentID → Appointments.AppointmentID	60,000	55,000	91.7%
Workbook Diagnoses AppointmentID → Appointments.AppointmentID	5,000	0	0.0%
Beds.RoomID → Rooms.RoomID	2,000	0	0.0%
Rooms.DepartmentID → Departments.DepartmentID	400	0	0.0%
Equipment.DepartmentID → Departments.DepartmentID	200	0	0.0%

Interpretation and limitations

The workbook has 5,000 appointment rows while billing and diagnosis CSVs reference appointment IDs from a 60,000-ID range; 55,000 rows in each child table fail to match the supplied appointment master. Feedback has 19,308 unmatched appointment references. The department master has valid IDs, but 4,744 appointment rows pair a facility ID inconsistent with the facility attached to their department. These patterns may indicate a truncated/mismatched appointment export or inconsistent synthetic data generation. No assumptions were made to repair these joins.

Bed status is Available for all 2,000 rows, so occupied-bed utilization cannot be calculated. Feedback numeric rating averages 2.99/5, but comment categories appear across all numeric ratings; sentiment cannot be inferred from the comment field. Billing and payment records are one-to-one by BillingID, but 11,887 payment amounts differ from billed amounts; finance status categories do not align cleanly enough to infer outstanding balances without a business rule.

Dashboard refinement recommendations

Build the dashboard with visible data-quality warnings. Use patient and appointment metrics only within their valid source tables. Keep billing, payments and monthly revenue summary as separate measures until reconciled. Suppress facility-level feedback attribution until appointment keys match. Do not display bed utilization as a KPI until occupied/available status values are meaningful. Include source date coverage, denominator notes, and unmatched-key counts in a Data Quality page.

Cross-track validation

Data Analytics → Data Science: share the appointment master coverage warning and confirm the modelling team's appointment population and target label source. Agree which appointment file is authoritative; reconcile ID ranges and row counts; document excluded unmatched records; then retest model feature/target extraction. This is a concrete dependency identified from the files; partner confirmation and retest evidence must be recorded after the actual collaboration.

Week 8 actions

1) Obtain authoritative full appointment master and resolve 55,000 unmatched billing/diagnosis references. 2) Resolve 19,308 unmatched feedback references. 3) Correct or explain 4,744 facility/department conflicts using approved master data. 4) Reconcile billed, paid, outstanding and monthly revenue definitions with finance. 5) Validate bed-status snapshot. 6) Clarify feedback comment/rating semantics. 7) Re-run KPI tests and Power BI reconciliation, archive screenshots and retest evidence.

Conclusion

The uploaded data now supports a substantive Week 7 testing report, a dataset inventory, foreign-key validation, descriptive analysis and a prioritised issue register. The most important result is not a facility ranking: it is that key source tables are currently inconsistent and must be reconciled before integrated operational, financial or patient-experience conclusions can be trusted.